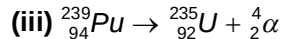
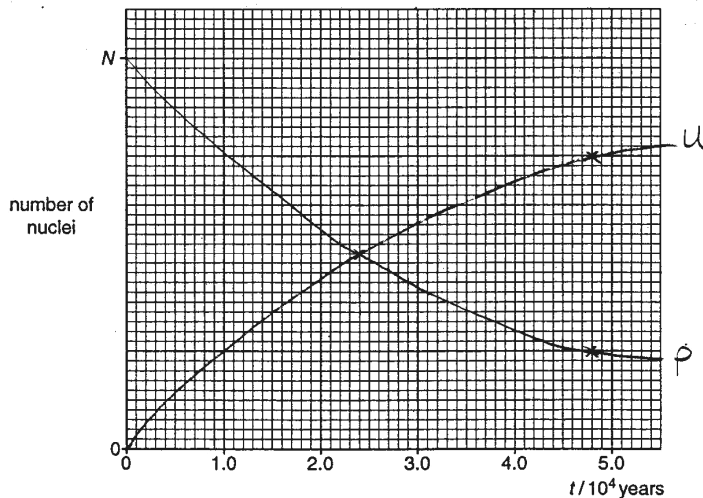


- 5 (a) (i) Isotopes are atoms/nuclides with the same number of protons but different number of neutrons. Here, the isotope uranium-X means that the proton number is the same, but the nucleon number is different for the uranium nucleus.
- (ii) Half-life is the average time taken for the number of radioactive nuclei to decay to half of its original number.

Note: No marks awarded for answers such as substance, concentration, atoms, mass or nuclides.



(b)



Note: Candidates must read the question and follow the instructions to label the graphs!

(c) When Pu decay, both U and α emitted will be moving in order to conserve momentum. Hence, some of the energy released may be in the form of kinetic energy of U, rather than just kinetic energy of α only. Furthermore, some of the energy released may manifest as gamma radiation.

6 (a) When r is between 0.9×10^{-3} m and 1.5×10^{-3} m, Q increases at an increasing rate